

Week 2

1 DINO and Self-Supervised Learning

In Week 1, we used a pretrained DINOv2 model to turn galaxy images into numerical representations. Week 2 explores how a model can learn useful visual features in the first place, without being given a class label for every image. This approach is called self-supervised learning because the training signal is created from the images themselves.

2 Load the CIFAR-10 Dataset

The notebook uses CIFAR-10, a dataset of small colour images belonging to categories such as cars, birds, cats, and ships. For this experiment, a seeded subset of 5,000 training images is selected. Using a subset keeps the experiment small enough to run as an educational implementation while making the selection reproducible.

3 Multi-Crop Augmentation

Instead of showing the model only one version of an image, the notebook creates multiple views of the same image. Two global crops show larger portions of the image at 32×32 pixels, while two local crops show smaller regions at 16×16 pixels. The views are different, but they come from the same original image.

Original image $\rightarrow$ global views + local views

4 The Tiny Vision Transformer

The notebook builds a small Vision Transformer (TinyViT) rather than loading a large pretrained network. The model divides each crop into patches and processes their representations with Transformer layers. In this implementation, the patch size is 8 pixels, the embedding dimension is 192, and the network has four Transformer layers. The smaller model makes the training process easier to inspect.

5 The Student and Teacher Networks

DINO uses two networks with the same basic architecture: a student and a teacher. The student is trained using a loss function. The teacher provides target probability distributions based on other views of the same image. The teacher is not updated by ordinary backpropagation; instead, its parameters follow a moving average of the student's parameters.

6 Understanding the DINO Loss

The student and teacher produce output distributions. The loss encourages the student to match the teacher's output for different views of the same image. The implementation compares teacher global views with student views while excluding the matching global-view pair. This encourages the model to learn representations that remain useful across changes in crop and appearance.

7 EMA: The Momentum Teacher

EMA means exponential moving average. After a student update, the teacher parameters are moved slightly toward the student parameters. The teacher therefore changes more smoothly instead of copying every student update directly. This helps provide more stable targets during training.

8 The Training Loop

For each batch, the notebook creates multiple crops, passes them through the student and teacher, computes the DINO loss, updates the student with gradient descent, and updates the teacher using EMA. The training configuration uses 5,000 CIFAR-10 images and runs for 10 epochs. The notebook records the training loss so that its behaviour can be plotted.

9 Inspect the Learned Representations

After training, the teacher network is used to extract CLS embeddings from 1,000 CIFAR-10 test images. A CLS embedding is one vector intended to summarize an image. These vectors can be compared numerically, just as the galaxy embeddings were compared in Week 1.

10 PCA and t-SNE Visualizations

An embedding can contain many dimensions, which makes it difficult to plot directly. PCA and t-SNE reduce the vectors to two dimensions for visualization. Each point represents an image. The plots can help reveal whether images with similar visual features tend to appear near one another, but a two-dimensional plot does not prove that the model has learned perfect categories.

11 Attention Map

The notebook visualizes attention from a trained Transformer layer. Attention values indicate which tokens interact strongly with other tokens inside that layer. Displaying the attention over the image gives a rough visual clue about the regions influencing the CLS token. Attention maps should be treated as diagnostic visualizations, not as complete explanations of the model.

12 EMA Ablation

An ablation compares the main setup with a modified setup to understand the contribution of one design choice. The notebook includes a small comparison involving the EMA teacher. This part uses a smaller subset and fewer epochs than the main run, so it should be interpreted as an educational comparison rather than a definitive scientific conclusion.

13 Results and Limitations

The notebook is designed to produce a training-loss plot, PCA and t-SNE plots, an attention map, and an EMA comparison. The numerical outcomes must be read from the completed notebook run; they are not stated here because no verified run results were provided. The experiment also uses a small dataset, a compact model, and a limited number of epochs, so its representations may be less strong than those from a large-scale DINO model.

The Big Picture

The Week 2 pipeline can be summarized as:

CIFAR-10 image → multiple crops → student and teacher TinyViT → DINO loss and EMA
→ embeddings and visualizations

The main idea is that the model learns by comparing different views of the same image rather than depending on labels for every training example. Week 1 showed how to use DINOv2 embeddings; Week 2 builds a small DINO-style training process to explore one way such representations can be learned.